Global X Camera

STM32N657 / VD55G0 camera | Functional schematic index

Recovered from Flux on 2026-09-05. Editable source transcription; hardware signoff remains pending.

Global net labels connect sheets. Each physical component pin is explicit. Original reference designators are retained.

Read ../reconstruction/README.md for source reconciliation, assumptions, pin audits and ERC findings.

2 Battery latch and shutter

File: 02_latch_1.kicad_sch

3 Main regulated rails

File: 03_rails_1.kicad_sch

4 MCU I/O and analog power

File: 04_mcu_power_1.kicad_sch

5 MCU I/O and analog power — continued

File: 05_mcu_power_2.kicad_sch

6 MCU core supply

File: 06_mcu_core_1.kicad_sch

7 MCU grounds

File: 07_mcu_ground_1.kicad_sch

8 Clock, boot and debug

File: 08_control_1.kicad_sch

9 Camera power

File: 09_camera_power_1.kicad_sch

10 Camera and CSI-2

File: 10_camera_1.kicad_sch

11 Boot NOR flash

File: 11_boot_1.kicad_sch

12 Raw NAND storage

File: 12_nand_1.kicad_sch

13 USB-C and USB High-Speed

File: 13_usb_1.kicad_sch

14 RTC and orientation

File: 14_aux_1.kicad_sch

15 Unused MCU terminals

File: 15_unused_1.kicad_sch

16 Unused MCU terminals — continued

File: 16_unused_2.kicad_sch

ERC supply-source declarations: external inputs and post-passive power paths; these are not a power-design signoff.

BAT_PLUS

BAT_CELL1_TAP

USB_VBUS

SYS_IN

3V3

1V8

VCORE_0V81

GND

RECONSTRUCTED SOURCE — review notes and ERC report apply

Global X Camera		
Sheet: /		
File: global-x-camera.kicad_sch		
Title: Global X Camera — source reconstruction		
Size: A3	Date: 2026-09-05	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 1/16

Battery latch and shutter

External battery stack, power hold, shutter sense and status LED.



RECONSTRUCTED SOURCE — review notes and ERC report apply

Source recovery; electrical design review pending

Global X Camera

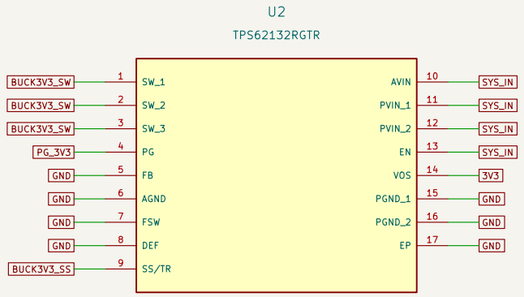
Sheet: /2 Battery latch and shutter/
File: 02_latch_1.kicad_sch

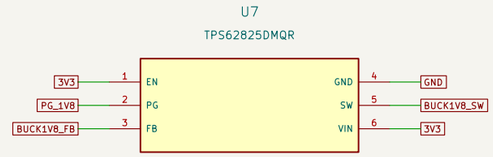
Title: Battery latch and shutter

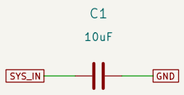
Size: A3	Date: 2026-09-05	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 2/16

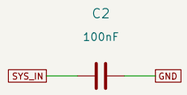
Main regulated rails

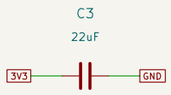
3V3 and 1V8 converters. Values and package choices recovered from Flux.



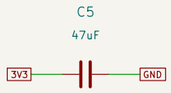








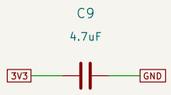


















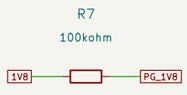












RECONSTRUCTED SOURCE — review notes and ERC report apply

Source recovery; electrical design review pending

Global X Camera

Sheet: /3 Main regulated rails/

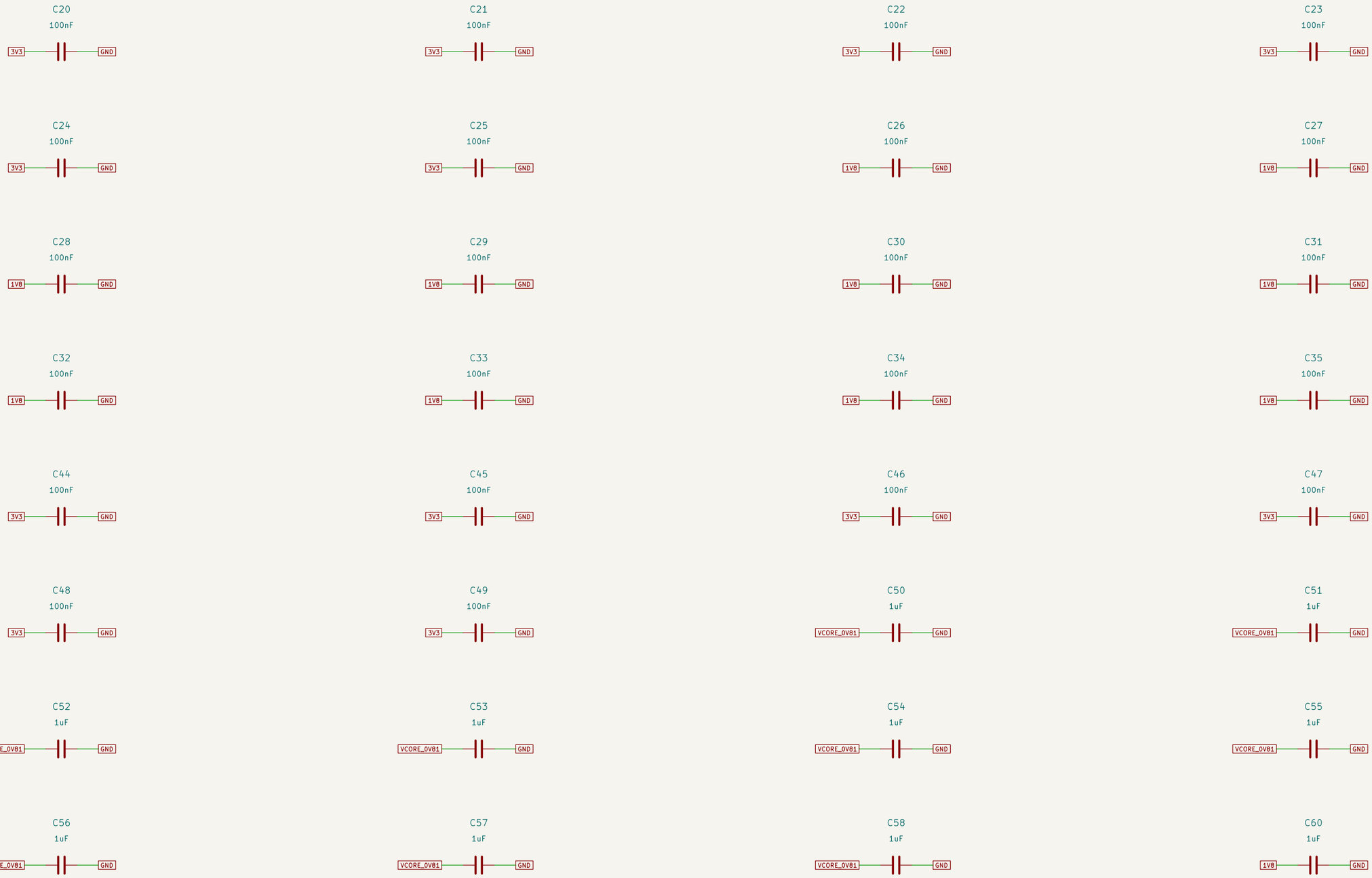
File: 03_rails_1.kicad_sch

Title: Main regulated rails

Size: A3	Date: 2026-09-05	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 3/16

MCU I/O and analog power

Individual supply balls and local bypass components; placement still to be reviewed.



RECONSTRUCTED SOURCE — review notes and ERC report apply

Source recovery; electrical design review pending

Global X Camera

Sheet: /4 MCU I/O and analog power/
File: 04_mcu_power_1.kicad_sch

Title: MCU I/O and analog power

Size: A3	Date: 2026-09-05	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 4/16

MCU I/O and analog power — continued

Individual supply balls and local bypass components; placement still to be reviewed.



RECONSTRUCTED SOURCE — review notes and ERC report apply

Source recovery; electrical design review pending

Global X Camera

Sheet: /5 MCU I/O and analog power — continued/
File: 05_mcu_power_2.kicad_sch

Title: MCU I/O and analog power — continued

Size: A3	Date: 2026-09-05	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 5/16

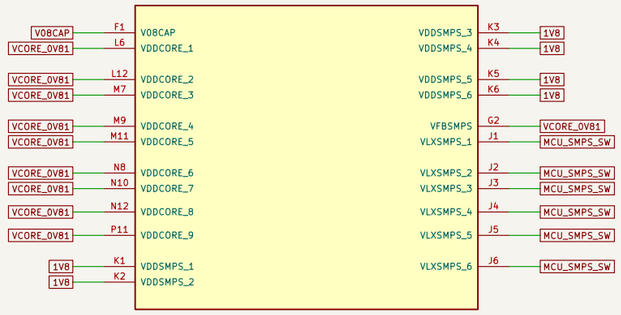
MCU core supply

Internal SMPS terminals and external core supply components.

Core supply and internal SMPS

U1B

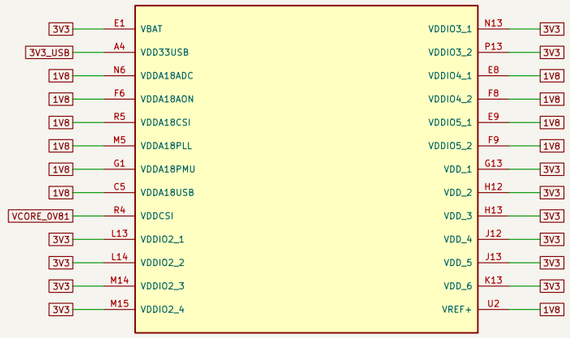
STM32N657X0H3Q



I/O and analog supplies

U1A

STM32N657X0H3Q



C69

1uF



R19

2ohm



C15

22uF



C16

22uF



C17

22uF



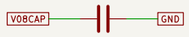
C18

22uF



C19

4.7uF



C59

2.2nF



L3

1uH



RECONSTRUCTED SOURCE — review notes and ERC report apply

Source recovery; electrical design review pending

Global X Camera

Sheet: /6 MCU core supply/

File: 06_mcu_core_1.kicad_sch

Title: MCU core supply

Size: A3	Date: 2026-09-05	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 6/16

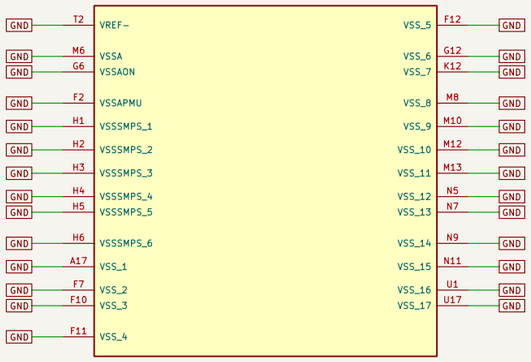
MCU grounds

Every ground ball remains individually numbered for package and netlist review.

MCU grounds

U1C

STM32N657X0H3Q



RECONSTRUCTED SOURCE — review notes and ERC report apply

Source recovery; electrical design review pending

Global X Camera

Sheet: /7 MCU grounds/

File: 07_mcu_ground_1.kicad_sch

Title: MCU grounds

Size: A3	Date: 2026−09−05	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 7/16

Clock, boot and debug

40 MHz oscillator, reset, boot straps, SWD and system controls. SW2/SW3 are DNP intent reconstruction.

Clock, boot, debug and system control

U1D
STM32N657X0H3Q



J2
TC2030-IDC-NL



C36
8.2pF



C37
8.2pF



C38
100nF



R8
10kohm



R9
10kohm



SW2
B3U-3000P



SW3
B3U-3000P



TP3
TP3



TP4
TP4



TP5
TP5



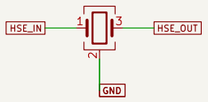
TP6
TP6



TP7
TP7



Y1
CX3225SA40000D0PTWCC



RECONSTRUCTED SOURCE — review notes and ERC report apply

Source recovery; electrical design review pending
Global X Camera

Sheet: /8 Clock, boot and debug/
File: 08_control_1.kicad_sch

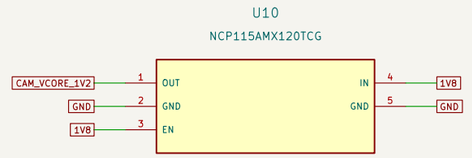
Title: Clock, boot and debug

Size: A3	Date: 2026-09-05	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 8/16

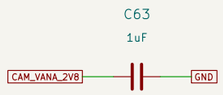
Camera power

1.2V core and switched 2.8V analog rails. Check firmware sequencing separately.

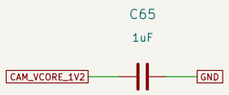






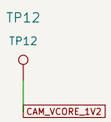












RECONSTRUCTED SOURCE — review notes and ERC report apply

Source recovery; electrical design review pending

Global X Camera

Sheet: /9 Camera power/
File: 09_camera_power_1.kicad_sch

Title: Camera power

Size: A3	Date: 2026-09-05	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 9/16

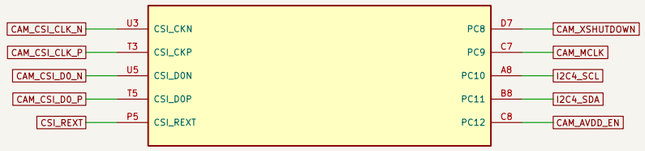
Camera and CSI-2

One CSI-2 data lane plus clock. CAM1 is an external module and is excluded from the host PCB.

Camera interface

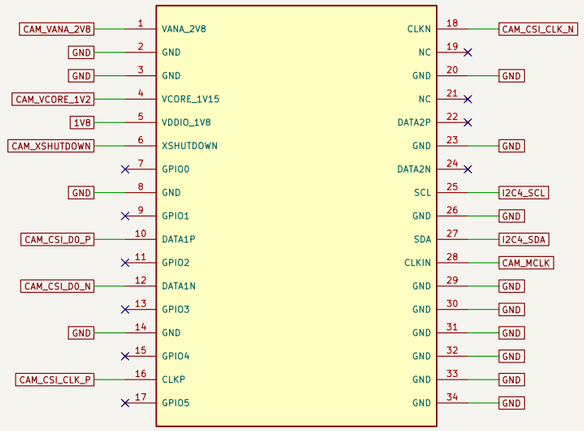
U1G

STM32N657X0H3Q



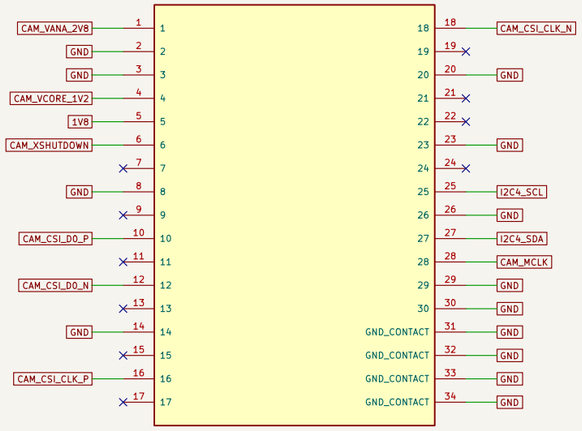
CAM1

CAM-5G0-079CLR



J3

BM28B0.6-30DS/2-0.35V(51)



C66

100nF



C70

1uF



R17

200ohm



R20

100kohm



R22

4.7kohm



R23

4.7kohm



TP13

TP13



RECONSTRUCTED SOURCE — review notes and ERC report apply

Source recovery: electrical design review pending

Global X Camera

Sheet: /10 Camera and CSI-2/

File: 10_camera_1.kicad_sch

Title: Camera and CSI-2

Size: A3	Date: 2026-09-05	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 10/16

Boot NOR flash

W25Q128JVPIQ, Quad-SPI boot. IQ variant has fixed QE; pins 3/7 are IO2/IO3.

Boot flash interface

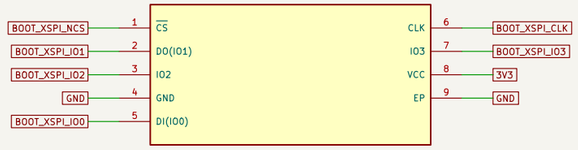
U1E

STM32N657X0H3Q



U5

W25Q128JVPIQ



C39

100nF



R13

10kohm



RECONSTRUCTED SOURCE — review notes and ERC report apply

Source recovery; electrical design review pending

Global X Camera

Sheet: /11 Boot NOR flash/

File: 11_boot_1.kicad_sch

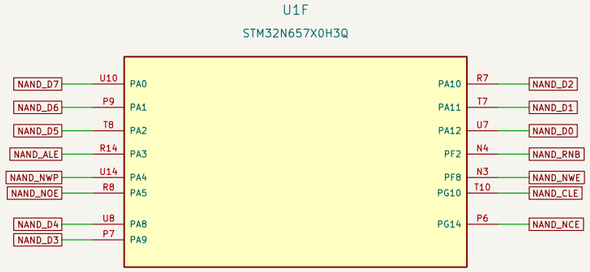
Title: Boot NOR flash

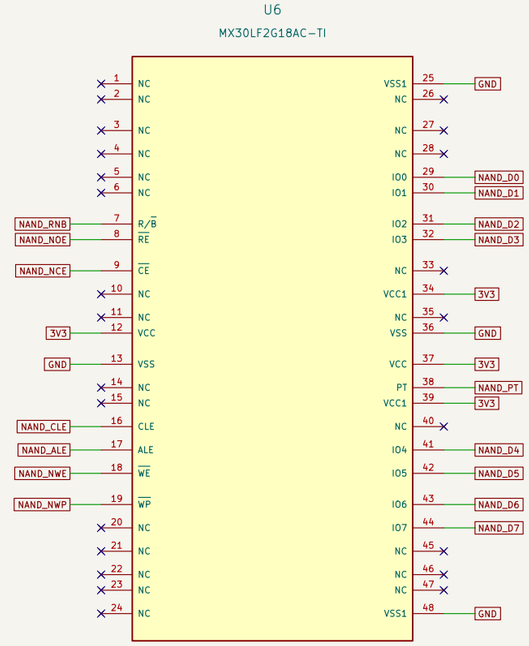
Size: A3	Date: 2026-09-05	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 11/16

Raw NAND storage

MX30LF2G18AC-TI, x8 parallel interface. Firmware ECC and timing review remains separate.

NAND interface





















RECONSTRUCTED SOURCE — review notes and ERC report apply

Source recovery; electrical design review pending

Global X Camera

Sheet: /12 Raw NAND storage/
File: 12_nand_1.kicad_sch

Title: Raw NAND storage

Size: A3	Date: 2026-09-05	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 12/16

USB-C and USB High-Speed

USB2 device interface, CC pull-downs, protection and USB-derived power.

USB interface

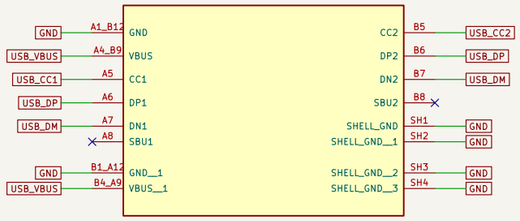
U1H
STM32N657X0H3Q



D1
USBLC6-2SC6



J1
USB4105-GF-A



U4
TLV70033DDCT



C13
1uF



C14
1uF



C43
4.7uF



D2
SS34



R10
5.1kohm



R11
5.1kohm



R18
200ohm



RECONSTRUCTED SOURCE — review notes and ERC report apply

Source recovery; electrical design review pending
Global X Camera

Sheet: /13 USB-C and USB High-Speed/		
File: 13_usb_1.kicad_sch		
Title: USB-C and USB High-Speed		
Size: A3	Date: 2026-09-05	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 13/16

RTC and orientation

Backup timekeeping, switched accelerometer supply and MCU interfaces.

RTC and accelerometer interface

U1I

STM32N657X0H3Q



U8

RV-3028-C8-32.768kHz-1PPM-TA-QC



U11

LIS2DW12TR



U12

TPS22918DBVT



C67

100nF



C68

100nF



C72

100nF



C73

1uF



C74

1uF



R24

10kohm



R25

10kohm



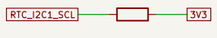
R26

1Mohm



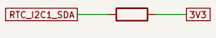
R28

4.7kohm



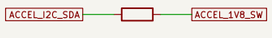
R29

4.7kohm



R30

10kohm



R31

10kohm



R32

100kohm



TP8

TP8



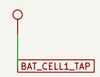
TP9

TP9



TP10

TP10



Source recovery; electrical design review pending

Global X Camera

Sheet: /14 RTC and orientation/

File: 14_aux_1.kicad_sch

Title: RTC and orientation

Size: A3	Date: 2026-09-05	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 14/16

RECONSTRUCTED SOURCE — review notes and ERC report apply

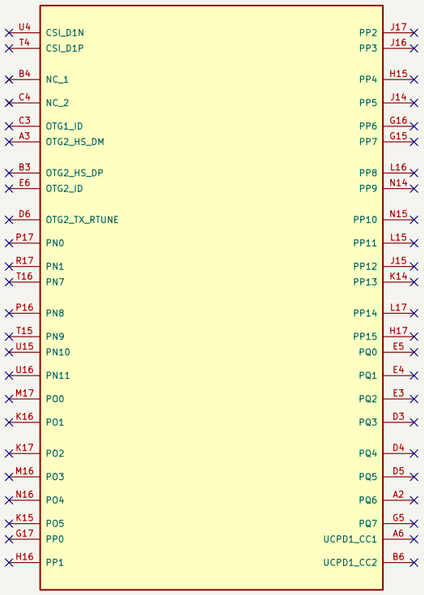
Unused MCU terminals

Explicit source no-connects, shown individually. No hidden or omitted MCU pins.

Unused GPIO N-Q and interfaces

U1L

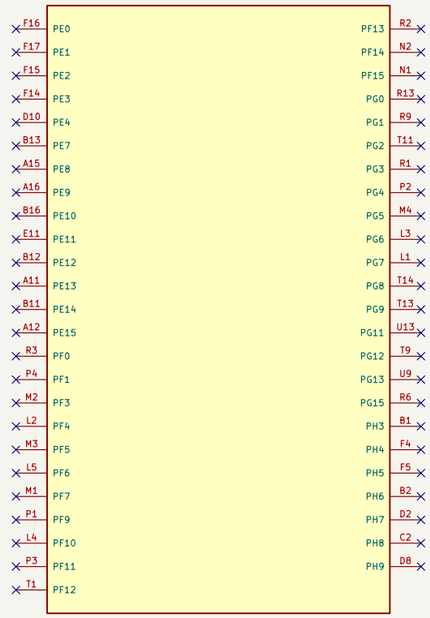
STM32N657X0H3Q



Unused GPIO E-M

U1K

STM32N657X0H3Q



RECONSTRUCTED SOURCE — review notes and ERC report apply

Source recovery; electrical design review pending

Global X Camera

Sheet: /15 Unused MCU terminals/

File: 15_unused_1.kicad_sch

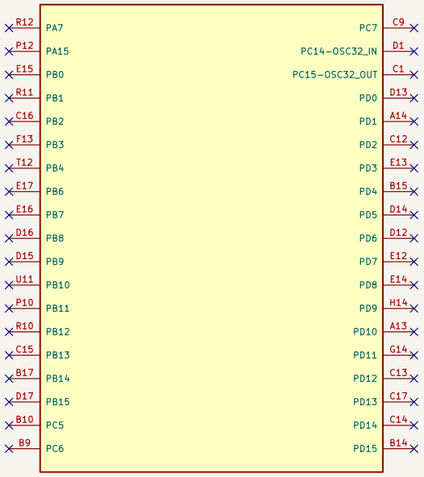
Title: Unused MCU terminals

Size: A3	Date: 2026-09-05	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 15/16

Unused MCU terminals — continued

Explicit source no-connects, shown individually. No hidden or omitted MCU pins.

Unused GPIO A–D



RECONSTRUCTED SOURCE — review notes and ERC report apply

Source recovery; electrical design review pending

Global X Camera

Sheet: /16 Unused MCU terminals — continued/
File: 16_unused_2.kicad_sch

Title: Unused MCU terminals — continued

Size: A3	Date: 2026-09-05	**Rev: A**
KiCad E.D.A. 10.0.6		Id: 16/16